

★ WEEK 12 · PART 2 of 3

The Term Structure and Bond Pricing

One Short Rate, a Yield for Every Maturity

Bond prices as expectations · the affine formula · yield-curve shapes

A single short-rate model implies an entire yield curve. We discount the random future rate into bond prices, turn prices into yields, and read the normal, inverted, and humped shapes the model can produce.

2

PART 2 · THE YIELD CURVE

From the Short Rate to the Curve

The term structure, bond prices as expectations of the discounted short rate, the Vasicek affine formula, yields from prices, and the shapes — normal, inverted, humped — one model can make.

From a Rate to a Whole Curve

Part 1 modeled the rate; Part 2 prices bonds

1

We have a short-rate model

Vasicek or CIR: a mean-reverting SDE for the instantaneous rate, with a few interpretable parameters.

2

But we quote yields, not the short rate

Markets trade bonds across maturities. We need the short rate to imply a yield for every term.

3

Discounting bridges the two

A bond's price is the expected discounted payoff at the random short rate. From prices come yields — and the whole curve.

Part 2 in one line: discount the random short rate into bond prices, then read off the yield curve.

The Term Structure of Rates

One short rate, a yield for every maturity

The term structure — the yield curve — is the set of yields across all maturities at one moment. A short-rate model must produce this entire curve from the single instantaneous rate and its dynamics.

What it is

- A yield for every maturity, from months to decades.
- Plotted, it is the familiar yield curve.
- It summarizes the price of money over time.

The modeling task

- Turn one short rate into yields at all terms.
- The dynamics of the rate shape the whole curve.
- A few parameters must generate the entire structure.

What a Yield Curve Is

Yields plotted against maturity

Reading it

- The horizontal axis is maturity — the term.
- The vertical axis is the yield for that term.
- Its slope and shape carry economic information.

Why it matters

- It prices every fixed-income instrument.
- Its slope reflects expectations and risk premia.
- Its inversions have famously preceded recessions.

From the Short Rate to Bond Prices

Discount the future at the random short rate

A zero-coupon bond pays one unit at maturity. Its price today is that payoff discounted along the future path of the short rate — but the rate is random, so the price is an EXPECTATION over all its possible paths.

The logic

- Discount the future payoff at the short rate as it evolves.
- The rate's path is uncertain, so average over paths.
- That average is the bond's fair price.

Why an expectation

- Each possible rate path gives a different discount.
- The price weighs them by their probabilities.
- This is where the model's dynamics enter.

The Bond Price as an Expectation

Averaging the discount over rate paths

$$P(0, T) = \mathbb{E} \left[\exp \left(- \int_0^T r_s ds \right) \right] \quad (\text{discount at the random short rate})$$

Reading it

- Inside: the discount factor over the whole path.
- The integral accumulates the short rate over time.
- The expectation averages across all possible paths.

What it needs

- The dynamics of the short rate — our SDE.
- For Vasicek and CIR, this has a closed form.
- So no simulation is needed for the price itself.

Why It Averages Over Paths

The rate is random, so the discount is too

The randomness

- The short rate follows a stochastic path.
- A high-rate path discounts the payoff heavily.
- A low-rate path discounts it lightly.

The resolution

- The price is the probability-weighted average.
- Mean reversion keeps the paths well-behaved.
- The model turns this average into a formula.

The Vasicek Bond Price

An affine formula in the short rate

$$\text{Vasicek: } P(t, T) = A(\tau) e^{-B(\tau) r_t}, \quad B(\tau) = \frac{1 - e^{-\kappa \tau}}{\kappa}, \quad \tau = T - t$$

The structure

- The price is a factor A times an exponential of the rate.
- The exponent is minus B(tau) times the current rate.
- A and B depend only on the maturity and parameters.

Why it is elegant

- No simulation — the expectation is solved exactly.
- The log-price is linear in the short rate (affine).
- CIR has a similar, if messier, closed form.

From Bond Prices to Yields

One short step to the curve

$$\text{yield from price: } y(0, T) = -\frac{\ln P(0, T)}{T}$$

The definition

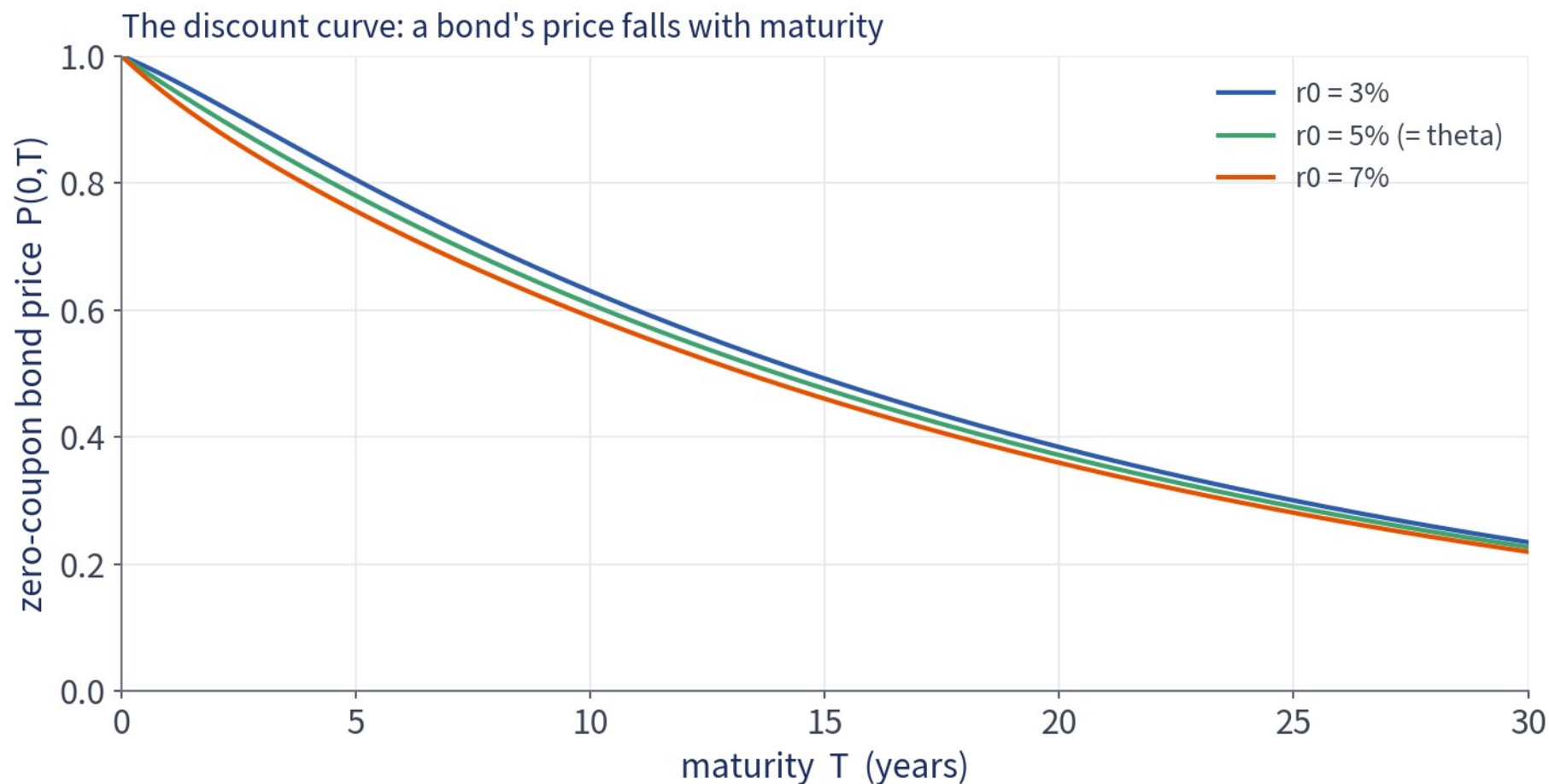
- The yield is minus the log-price divided by maturity.
- It is the constant rate that reproduces the price.
- Compute it at each maturity to trace the curve.

What you get

- A yield for every term — the whole curve.
- All from the one short rate and its parameters.
- Prices and yields are two views of the same thing.

The Discount Curve

The zero-coupon bond price falls with maturity: money further away is discounted more. A higher starting short rate discounts harder, pushing the whole price curve down.



Affine Term-Structure Models

Yields linear in the short rate

affine model: $y(0, T) = a(T) + b(T) r_0$ (the yield is linear in the short rate)

What 'affine' means

- Each yield is a linear function of the short rate.
- The intercept and slope depend on maturity.
- Vasicek and CIR are both affine models.

Why it is prized

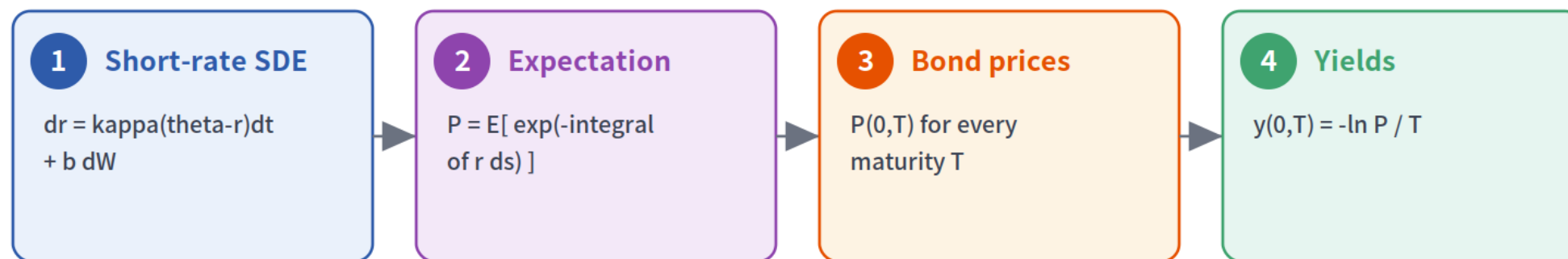
- Closed-form prices and yields — fast to compute.
- Easy to calibrate and to reason about.
- It is the workhorse class of rate models.

From One Short Rate to the Whole Curve

The short-rate SDE feeds an expectation of the discounted path, which gives bond prices at every maturity, which convert to yields. Fix the parameters and r_0 , and the model prints an entire curve.

From One Short Rate to the Whole Curve

The short-rate SDE feeds every step: expectation, then prices, then yields, then the curve.



The whole curve from a handful of parameters

Fix κ , θ , σ and the starting short rate r_0 , and the model prints a yield for every maturity - an entire term structure. In an AFFINE model (Vasicek, CIR), each yield is simply linear in r_0 .

One short rate today, a full curve of yields tomorrow.

What Shapes Can the Model Make?

Normal, inverted, and humped

A single short-rate model produces different curve shapes depending mainly on where the rate starts relative to its long-run level. From one model come upward, downward, and humped curves — the shapes seen in real markets.

The three basic shapes

- Normal: yields rise with maturity.
- Inverted: yields fall with maturity.
- Humped: yields peak at a middle maturity.

What drives them

- Mostly where today's rate sits versus the mean.
- Below the mean tends to give an upward curve.
- Above the mean tends to give an inverted one.

The Normal (Upward) Curve

The most common shape

When it appears

- The short rate is below its long-run level.
- The market expects rates to rise toward the mean.
- Longer bonds also carry a term premium.

How it reads

- Longer maturities yield more than shorter ones.
- It reflects a normal, expansionary environment.
- It is the shape most of the time.

The Inverted (Downward) Curve

The recession signal

When it appears

- The short rate is above its long-run level.
- The market expects rates to fall back down.
- Often tight policy today, easing expected ahead.

How it reads

- Longer maturities yield LESS than shorter ones.
- It has historically preceded recessions.
- A closely-watched macro warning sign.

The Humped Curve

A peak in the middle

What it is

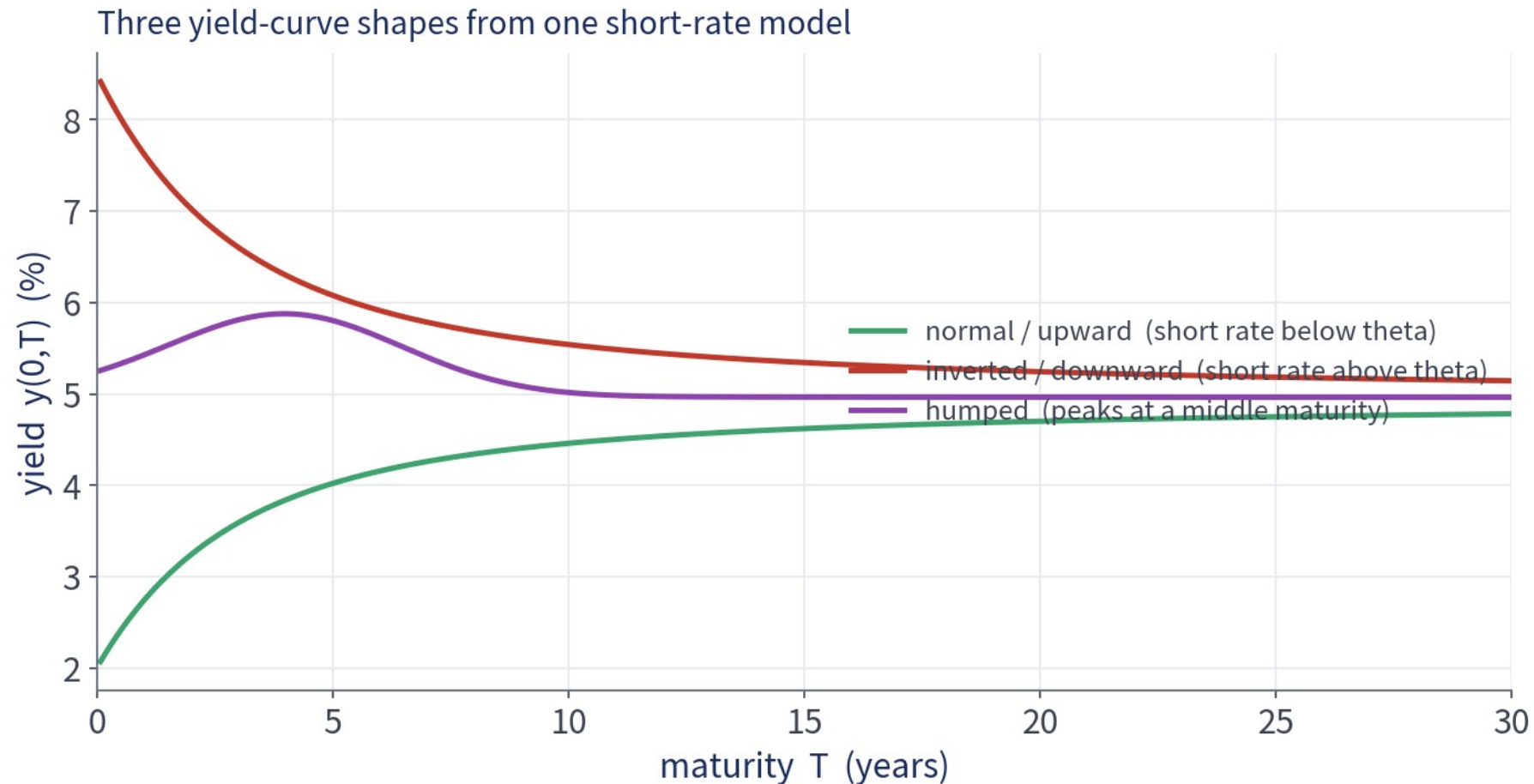
- Yields rise, peak at a medium term, then fall.
- A transitional shape between normal and inverted.
- It appears around turning points in policy.

Why the model makes it

- Mean reversion plus a term premium can peak mid-curve.
- Short and long yields converge to the long-run yield.
- Intermediate maturities can sit above both.

Yield-Curve Shapes

Normal (upward), inverted (downward), and humped curves — all produced by one short-rate model. The shape follows mainly from where today's short rate sits relative to its long-run level.



Today's Short Rate Sets the Shape

One parameter tilts the whole curve

Low starting rate

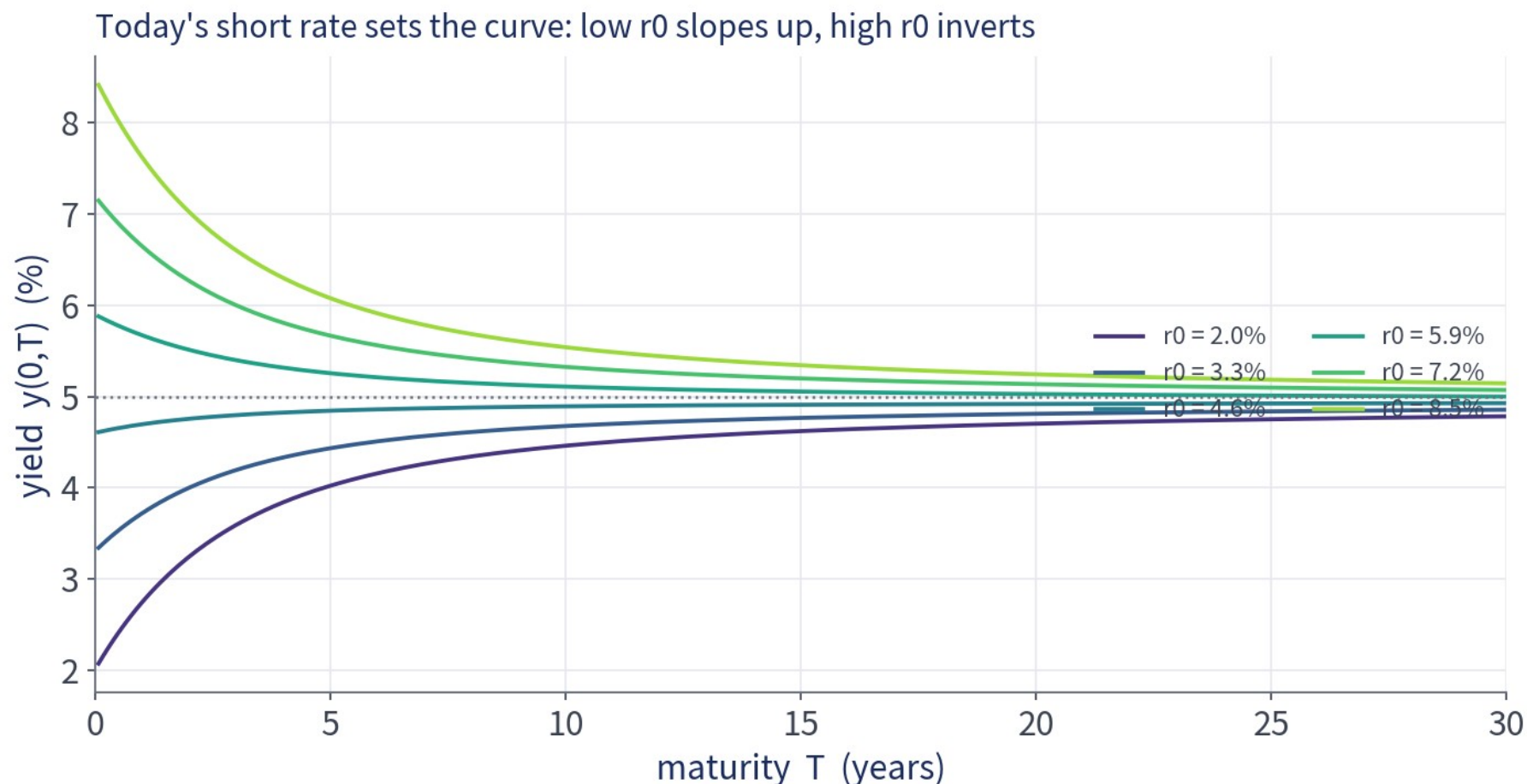
- Below the long-run level, rates are expected to climb.
- So the curve slopes upward.
- The classic normal shape.

High starting rate

- Above the long-run level, rates are expected to fall.
- So the curve slopes downward.
- The inverted shape.

The Short Rate Sets the Curve

Sweeping the starting short rate from low to high tilts the whole curve from upward-sloping to inverted. All curves converge to the same long-run yield — the reversion level shows through at the long end.



The Economic Reading

What the curve is telling you

The signal in the slope

- Upward: growth and gradually rising rates expected.
- Inverted: tight policy now, cuts expected ahead.
- The curve encodes the market's rate expectations.

A caution

- The model is a stylized lens, not the whole story.
- Real curves also reflect liquidity and supply effects.
- Read shapes as signals, not certainties.

What Part 3 Adds

From formulas to computation

Discretization

- Turn the SDEs into steps a computer can run.
- Euler-Maruyama, its pitfalls, and Milstein.
- The tools to simulate any short-rate model.

Calibration and pricing

- Fit the model's parameters to market data.
- Price a bond end-to-end by simulation.
- Cross-check simulation against the closed form.

Build a Yield Curve From a Model

Direct Claude Code to turn a rate into a curve

GOAL Use the Vasicek closed-form to build yield curves, and see how the starting short rate sets the shape.

[INPUT] Prompt · Claude Code

Write and run `yield_curve.py` that:

1. implements the Vasicek bond price
 $P(0,T) = A(T) \exp(-B(T) r_0)$ with
 $B(T) = (1 - e^{(-\kappa T)}) / \kappa$;
 2. converts prices to yields $y(T) = -\ln P / T$;
 3. plots the curve for several starting rates r_0 (below, at, and above θ);
 4. labels each curve normal, flat, or inverted.
- Show that low r_0 slopes up and high r_0 inverts, and that all curves meet at the long-run yield.

How to work with it

- The yield is linear in r_0 (affine).
- Low r_0 gives an upward curve.
- High r_0 gives an inverted curve.
- Long-maturity yields converge together.

Deliverable: `yield_curve.py` + Vasicek yields across maturities for several starting rates.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1	A zero-coupon bond price is the expectation of what?	The discounted short rate
Q2	In an affine model, how does a yield depend on the short rate?	Linearly
Q3	What mainly sets whether the curve is upward or inverted?	r_0 vs the long-run level

All three come from the pricing expectation, the affine form, and the starting short rate.

Three Misconceptions to Avoid

Where curve intuition slips

Common myths

- "You must simulate to price a Vasicek bond."
- "The curve shape is unrelated to today's rate."
- "An inverted curve guarantees a recession."

The reality

- Vasicek and CIR have closed-form bond prices.
- The starting short rate largely sets the shape.
- Inversion is a signal with a good record, not a certainty.

Three Things to Carry Forward

Part 2 in one screen

1

A bond price is an expectation

It is the expected discount of the payoff along the random path of the short rate. For Vasicek and CIR this expectation solves in closed form — no simulation needed to price.

2

Affine models give the whole curve

Each yield is linear in the short rate, so fixing the parameters and today's rate prints a yield for every maturity — an entire term structure from a handful of numbers.

3

One model, many shapes

Normal, inverted, and humped curves all come from a single short-rate model, driven mainly by where the rate starts relative to its long-run level.

Next — Part 3: discretizing the SDEs, calibrating to data, and pricing a bond by simulation.